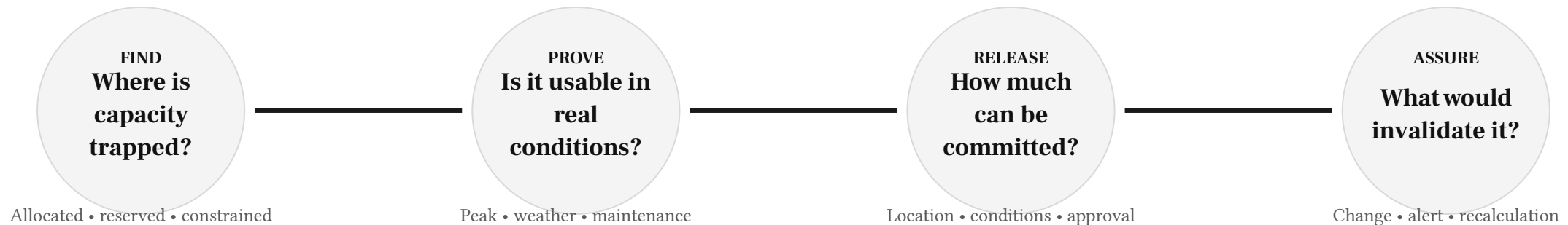


Turn hidden headroom into capacity the owner can safely sell

We help a data-centre owner safely sell capacity they already have but cannot confidently offer to a customer today.



Find the hidden capacity.

Prove it is usable.

Release it safely.

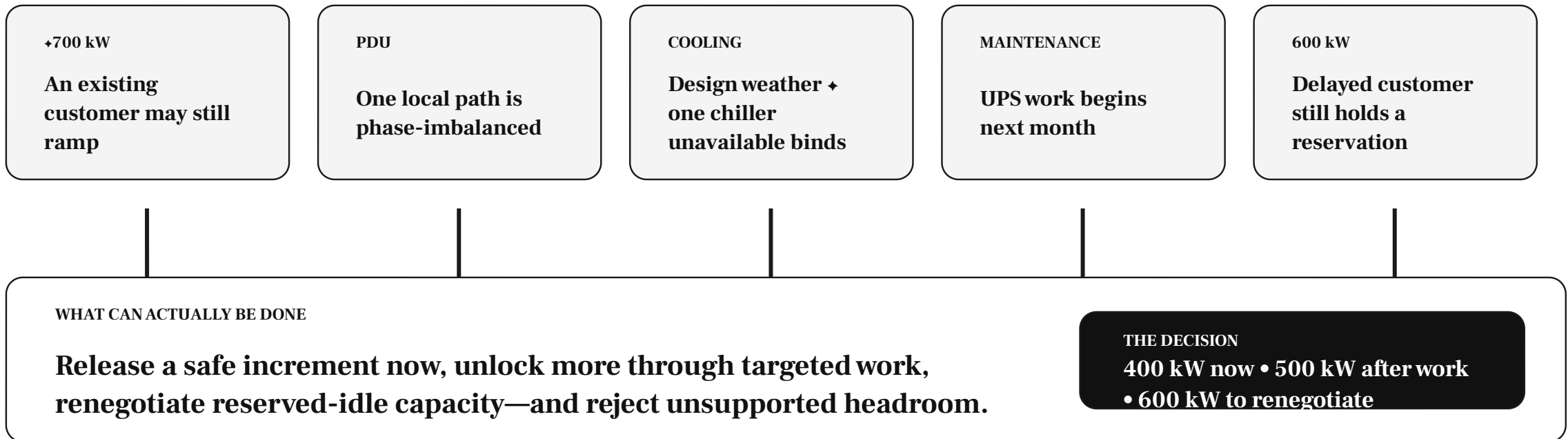
Keep proving it.

For operations, commercial, capacity and facilities-engineering teams deciding how much load can be contracted, where and under which conditions.

The hall appears to have 1.9 MW free—but most of it cannot be sold as-is

8 MW OPERATING HALL

7.2 MW is allocated; only 5.3 MW appears at coincident peak.



A commercial commitment must survive contractual ramps, local constraints, maintenance, design weather and agreed resilience states.

A location-specific capacity release decision—not another utilisation dashboard

1.1 MW RECOVERABLE

Hall A can release capacity in three controlled steps.

Release now	400 kW supported by current evidence
Release after action	300 kW after phase rebalancing
Commercial recovery	400 kW held by an expired reservation
Governing condition	Cooling binds above 6.4 MW in design weather with Chiller 3 unavailable
Validity window	Valid until UPS maintenance begins on 14 September

CAPACITY STATUS		
	Now	Evidence supports commitment
	With action	A named change unlocks it
	Commercial	Held by reservation or contract
	Conditional	Valid within explicit limits
	Not releasable	Hard engineering or contract limit
	Recalculate	A material change invalidates it

Engineers approve technical release. Authorised commercial owners approve the customer commitment; the system keeps the evidence and validity conditions visible.

Release the cheapest safe increment first—and keep every MW defensible

TODAY

Installed, allocated, measured and safely releasable capacity are treated as competing numbers.

- Current under-use looks sellable
- Site totals hide local constraints
- Normal weather becomes the planning case
- Recovery options compete informally
- A study becomes stale after approval

WITH CAPACITY RECOVERY

Commercial inventory is tied to location, constraints, scenarios and engineering approval.

- ✓ Respect customer ramps, reservations and future demand
- ✓ Find the breaker, PDU, phase, UPS or cooling zone that binds
- ✓ Test design weather, maintenance and approved contingencies
- ✓ Rank MW unlocked by cost, time, operational risk and approval need
- ✓ Recalculate when contracts, topology, equipment or conditions change

Practical value: more sellable MW, faster customer placement, deferred capex, recovered reservations and protected resilience.

Join engineering limits, live demand and contractual rights in one capacity model

THE DATA WE NEED

1	Physical paths	Electrical + cooling topology • ratings • settings
2	Actual behaviour	EPMS/BMS/DCIM • branch and zone telemetry
3	Asset states	Maintenance • degradation • availability
4	Customer rights	Allocations • ramps • reservations • expiry
5	Future demand	Sales pipeline • deployment • density profile
6	Assurance cases	Peak • weather • maintenance • contingency

HOW THE ANSWER IS BUILT

- 1 Reconcile capacity truth**
Join installed, allocated, reserved and measured capacity
- 2 Find the binding constraint**
Calculate power and cooling limits at every relevant node
- 3 Stress the proposed release**
Test ramp, peak, design weather, maintenance and failure states
- 4 Recommend and keep revalidating**
Release, recover or reject; monitor every invalidation trigger

ML forecasts demand inside the envelope. Deterministic engineering rules define the safe limit. Governed workflow routes the approval.

What must be true before we test the use case

Start with one constrained hall and one real proposed deployment—not a site-wide theoretical capacity study.

✓	Validated topology	Engineers can confirm electrical and cooling paths to proposed racks
✓	Granular telemetry	Sufficient history below site level, with known quality and states
✓	Commercial truth	Allocations, ramp rights, reservations and a real buyer or deployment
✓	Scenario definition	Maintenance, weather, peak and contingency states are agreed
✓	Named decision owners	Electrical, mechanical and commercial reviewers can act

If topology, cooling, contractual rights or engineering review are missing, position this as capacity-evidence readiness—not capacity release.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
● Rebuild how power and cooling reach the rack	Topology reconstruction	Convert drawings, registers and monitoring points into a dependency graph
● Join meters, assets, customers and contracts	Entity resolution	Connect physical assets, telemetry, allocations and rights
● Calculate the safe engineering limit	Deterministic constraint model	Apply ratings, derating, redundancy and cooling delivery at each node
● Forecast future demand and coincidence	Hierarchical forecasting	Estimate customer, circuit, zone and site demand with uncertainty
● Test every important operating condition	Scenario engine	Simulate peak, weather, maintenance, degradation and failure
● Find the best placement or recovery action	Constrained optimisation	Rank feasible locations and actions by MW, cost, time and risk

Control boundary: no autonomous setpoint, protection, topology, cooling, allocation or contract change—and no technical or commercial approval.